

A man wearing a blue work shirt with a 'pdg' logo and a white hairnet is focused on his work. He is wearing earbuds and looking down at a large, reflective metal plate with many circular holes. The background is a dark, industrial setting with a clock visible on the wall.

INFORM

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INNOVATION IS OUR KEY TO SUCCESS.

Dear Readers,

Our moulds enjoy an excellent reputation in the industry. This is not our own assessment, but the result of numerous discussions with customers and specialists in the field. In other words, we could lean back and rest on our laurels. But this would be unlike us. We're self-critical and keep analyzing our products time and again, verify that they're still competitive, technically at the latest level, or that they meet the ever higher expectations as regards minimum energy consumption.

We regard this task as the most important one in our entire enterprise. Innovation is a process in which viable, useful solutions are developed from ideas. Many of our customers appreciate this attitude and share their experiences with us.

We're also engaged in the development of suitable materials. In cooperation with well-known producers of plastics we examine and test new materials. We then use the test data and results in the design of our moulds and pass them on to the bottle producers who, at the end of the value creation chain, want to deliver good quality.

To be innovative is not only extremely exciting, it's also very challenging. Hereby the financial aspect must play more than just a background role. Our maxim is that we don't invent anything that can't be sold.

We speak of an innovation if all aforementioned aspects are in a reasonable balance. It's the only way for the long-term success of our enterprise, seeing that markets and products are subject to ever shorter cycles.

Yours, Otto Hofstetter.



Cover.

850 million preforms and 45 million bottles are produced annually by PDG Plastiques in the French town of Malesherbes.

The "rien n'est impossible" mentality of the innovative employees ensures that the enterprise is a sought-after partner of the foodstuffs industry.

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GROWING WITH FLOWER- POTS.

Soparco produces about **one billion flowerpots** each year. From its domicile in the French town of Condé-sur-Huisne the enterprise supplies horticulturists and tree nurseries all over the world. **Philippe Cohu, CEO of the family-owned business**, gives us an insight into his company.

With its coloured flowerpots Soparco satisfies an important customer requirement.

Mr Cohu, Condé-sur-Huisne isn't what one would call a well-known location of industry and commerce. What is it that brought Soparco to this place?

This is an interesting story. Have you got time? My father-in-law, Maxime Laguerre, laid the foundation stone for today's enterprise in 1970. He was a great innovator. He lived and worked in Paris at that time and had secured patents for many innovations. Two of these were revolutionary and very important for Soparco as it presents itself now: the "Plasti-Lien", a forerunner of today's cable ties, and a plastic clip for hanging up hand towels or closing bags. It is with these two articles that he made the breakthrough in the plastics production. He decided to enter this field personally as a producer. And he looked for a suitable location outside Paris. He found what he wanted in Condé-sur-Huisne.

**"QUITE CLEARLY SOPARCO'S
STRENGTH IS THE
MULTITUDE OF PRODUCTS FOR
NUMEROUS NICHES."**

Philippe Cohu, CEO

When did the specialization in articles for horticulturists and tree nurseries happen?

When his first machine started to produce plastic ties, he asked himself whether or not he should also offer them to horticulturists and nurseries to tie up plants and young trees. The idea worked. He got to know the branch of the industry at a time when, if at all, clay pots were used. As an innovator and specialist in the processing of plastics he resolved to make pots of polyethylene. It was a great idea! Soparco became one of Europe's most important producers of flowerpots within just a few months.

What do the figures look like today?

Soparco has two locations. Here in the Perche, as this region is called, is the main works. A subsidiary is in Chaingy near Orléans. We offer a secure job to about two hundred employees and operate fifty injection moulding machines. We also run four thermo-forming systems and six vacuum-forming packaging machines for polystyrene. We process a total of 14,000 tons of raw material to produce a billion pots and achieve sales to the tune of 35 million euros.

Which ones are your key products?

We've split up the markets. I believe that the added value for the customer increases the better our products are matched

to suit the different requirements. This is why we offer a very wide spectrum of products. To this extent Soparco doesn't really have a star. It's the variety of products with which we successfully cover numerous niche markets in France and abroad. The most remote customer is a horticulturist in Australia. For budget products with volumes from three to ten litres we use thermo-forming because the price is the most important criterion in this segment. Injection moulding, however, clearly is and remains our first domain where high-grade quality products are concerned.

We see enormous storehouses on your site. How come you have such large stocks?

We keep approx. 1,500 articles in store. This enables us to deliver 90 to 95 per cent of our products within two to three days. Otherwise our multitude of different products would prevent quick supplies. What's more, garden plants and young trees are a very seasonal business. Without stocks we would need much more production equipment to be able to satisfy the demand in peak periods. This would in turn lead to unjustifiable overcapacities during the "normal" months.

Which ones are your main markets?

France is our number one market, even if it hardly grows any more. Soparco holds a very strong position in its home market, and a dominating share which is very difficult to expand. Looking at our country's economic situation, the maxim is rather "to hold the market share". We see growth potential in exports, however.

The plastics industry hopes for new impulses from bio-based raw materials. What do things look like in this respect for Soparco?

We offer a complete bio line. The raw material for this was developed by us in cooperation with a lab in France, and it displays a very positive eco-balance. In the choice of materials we took care to avoid products which might also be used as foodstuffs as in the case of some other bio-products. Our raw material consists in the main of wood and wheat fibres, whereby we use the chaff rather than the grain. Elaborate tests have resulted in a bio-material which is suitable for both injection moulding and vacuum forming, subject to minor adjustments in production.

Is this the future for Soparco?

Not alone. In our opinion this is a niche market. Right now these pots of bio-materials are three to four times more expensive than conventional plastic pots. They sell well mainly in Switzerland and Austria, and in smaller quantities in France. Strangely

"WE DELIVER
ALMOST EACH ONE OF OUR
1,500 ARTICLES WITHIN
A MAXIMUM OF THREE DAYS."

Philippe Cohu, CEO



Soparco produces around the clock and on nearly 365 days per year.

enough, there's hardly any demand for bio-pots in Germany, where major efforts are normally made when it comes to matters of ecology. The Germans believe more in recycling, which is also a valid approach.

How important is recycled material in your production?

Unlike in the packaging of foodstuffs, there are no specific rules for the production of flowerpots. We're very careful all the same and use only colours and materials which meet the REACH regulations on chemicals of the EU. For the standard colours of black, terra-cotta and dark green we use nothing but recycled material. The ratio for bright and strong colours is fifty-fifty. I estimate that the portion of recycled material over our entire production is on average nearly 85 per cent.



Colours dominate your spectrum of products. Is this the trend of the time?

I'm not exaggerating by saying that Soparco is the initiator of coloured flowerpots. We made the first pots in colour already in 1988. Sales were rather sluggish at the outset. The market was ready a few years later, however, and the demand kept increasing steadily. Horticulturists were primarily the ones who bought our coloured pots, since their customers are glad not having to repot the plant into a decorative receptacle after the purchase. In the meantime, nurseries have also discovered advantages. They use the coloured pots to signal the clients the colour of the flowers, which normally becomes evident later on only. Other nursery people allocate the plants to price categories and differentiate between these with colours.

What support do you offer your customers in the layout of their range?

We rely very much on consultancy and share our findings with our customers. We also offer a wide range of extra articles for flowerpots such as handles, protective grids, hanging baskets and so on. This leads to a whole system which brings a nursery considerable extra business. We also give advice when it comes to colours and shapes and deal with issues of logistics.

**"TOGETHER WITH
OTTO HOFSTETTER AG WE HOLD
OUR POSITION IN THE MARKET."**

Philippe Cohu, CEO

Soparco's slogan is: Competence and innovation. Where does the innovation come from?

There are various sources. On the one hand, all our salespeople return with information and ideas from their visits to clients. Sometimes a customer comes up with a very special question to which he wants an answer. I, personally, am very open to new things and interested in changes concerning our branch of the industry. Perhaps I'm infected by the spirit of our company founder Maxime Laguerre. Some impulses come also from team discussions with the experts of Otto Hofstetter AG.



Soparco produces containers and accessories for horticulturists, flower growers and tree nurseries. The French leader in its field is domiciled in Condé-sur-Huisne and has a second production facility in Chaingy. www.soparco.com

To what extent do you get support from your mouldmaker?

Otto Hofstetter AG is our sparring partner whenever we have an idea for a big project which might be trendsetting and hold much commercial potential. Usually the Hofstetter management is ready without restrictions and in the spirit of a partnership to help bearing the risks of a new development. Where niche products and small series are concerned I try to cope with a French partner.

What do you appreciate in your Swiss partner?

We in the European neighbour countries regard Switzerland as an expensive place with quite high production costs. Otto Hofstetter AG, however, has much talent in offering an ab-



Ever more innovation keeps coming from Soparco thanks to inputs from many sides and the inventiveness of the company founder Maxime Laguerre.

solutely competitive technology. And it has been this way for decades. Apart from this, the specialists of Otto Hofstetter AG are always available to us. Solutions are sought unbureaucratically and quickly if, rarely, something doesn't work smoothly or isn't yet to our entire satisfaction. The cooperation is rather a team effort for which I'm very grateful.

If you had a wish to Otto Hofstetter AG, what would it be?

The prospects would be very good for us if Otto Hofstetter AG would stay where it is already now: at the forefront of innovation and technology, and in the lead when it comes to service. If the company keeps supporting us customers with these abilities, we also can hold our position in the market.

What does Soparco's future look like?

This is a delicate question. The economic situation in Europe is currently somewhat tight. Growth is nearly zero and all economies restrict spending. We at Soparco concentrate on stabilizing our currently solid basis with new products, a positive cash flow and a fair margin. So far we're quite successful in this. Our machinery is between four and five years old and thereby newer than the average in our trade. What's more, our injection moulds are also in perfect condition. Thanks to our know-how and our technical experience we're able to enter other fields. My philosophy is that no crisis lasts forever and that we will be fully ready when the economic motor starts turning again.

Mr Cohu, thank you very much for this discussion. We wish you every success and that you may soon experience a new spring with Soparco.

THE IDEA MAKES THE DIFFERENCE.

The application of high-tech can pay dividends even in such simple things as flowerpots. A new approach reduces the consumption of raw material and minimizes the cycle time without impairing the high product quality. The novel technology has been developed by Soparco and Otto Hofstetter AG.





Hardly any amateur gardener anywhere in the world is aware of the ingenious technology and know-how behind a flowerpot of plastic. Unless he/she is on the payroll of a company making flowerpots or moulds for flowerpots, of course. Like the employees of Soparco and Otto Hofstetter AG, for example. In a joint effort the two enterprises have developed a new technology. Compared with the common plastic flowerpots, the ones of the SX series, as they are referred to by Soparco, are lighter, easier on resources and produced faster.

Start at several points.

The most striking aspect is the clearly lighter weight of the flowerpot. In spite of this the pot meets the most varied pre-requisites for the plants to flourish. Apart from watering and draining factors, air circulation and soil contact, the design satisfies the high requirements also when it comes to the pot's stability. But why are the strength and weight of injection-moulded flowerpots a topic at all? It's entirely clear to the experts in the field: because the economical and thereby sustained use of raw material plays an ever more important role. The weight optimization is achieved with thinner walls of the pot. This leads not only to new challenges as regards the product's stability, but also to different demands on the moulding process. With all these points in mind, the mould-makers of Otto Hofstetter AG decided to resort to multi-point gating in the development of the SX range of flowerpots.

-20%

**MULTI-POINT GATING SAVES
20 PER CENT OF RAW
MATERIAL ON AVERAGE.**

Markedly reduced weight.

Multi-point gating shortens the flow paths in the mould cavity. The reduced distance from the gate to the farthest point at the product enables thinner wall sections. This brings down the pot weight and saves raw material. The specialists of Otto Hofstetter AG reckon with 20 per cent on average. This means that e.g. a 10-litre flowerpot of the SX series of Soparco is approx. 40 grams lighter than a conventionally moulded pot of the same size.

Shorter cycle time.

The parameters of the moulding process vary from those needed with one central gate. The thinner walls require less hold pressure and residual cooling time. The experts improved and optimized the moulding parameters in detail to the extent that an improvement of the cycle time of 1.5 seconds results.

**IN THE CASE OF A 10-LITRE
POT THE CYCLE TIME IS
SHORTENED BY 1.5 SECONDS.**

Design brings stability.

The design is also affected by the multi-point gating technique. The question of how much wall section is needed in which location of a product gets an entirely new meaning. Mould filling must take place in a much more defined way. Compared with one-point gating, there are much fewer possibilities of influencing injection. To take advantage of the new method it was necessary to take some completely new design approaches. Many modifications are hardly visible, but can have a decisive effect on the optimal functioning of the mould. One can no longer speak of a constant wall thickness, for example, because this parameter varies over

the complete flowerpot. Entirely new solution approaches had to be found to guarantee the properties required for perfect plant growth. This is how the new base design resembling a daisy materialized, among other characteristics. A detail which was accepted with pleasure by Philippe Cohu, both amateur gardener and President of Soparco.

Production follows.

The concept of the SX series imposed new demands also on the ejection of the pots. The stripper ring with the traditional short stroke was no longer sufficient to cope with the new properties of the product in the mould. This called for further supporting measures. The implementation of the new mould technology in production was a challenge also to Soparco. However, thanks to the vast experience of the company's team in injection moulding and the handling of flowerpots, the new moulds for the SX series were integrated without problems.

Yet more potential.

The multi-point gating technique was revolutionary for the realization of the SX series of Soparco, but can also be applied to other plant pots and indeed other products. The developers of Otto Hofstetter AG are confident that there is much room yet for further profitable optimizations.



The novel base geometry in the shape of a daisy provides the flowerpots of the SX series of Soparco with the necessary stability in spite of the markedly reduced pot weight.

IN SHAPE FOR THE FUTURE.

PDG Plastiques ranks among the major producers of PET preforms and bottles in France. With much flexibility and innovative genius, the **family enterprise** established in Malesherbes about two hours by car south of Paris has secured itself a **powerful position** for the future. **General Manager François Desfretier** allows Inform to have an insight into the company and its success.

Mr Desfretier, what does PDG stand for?

It's an abbreviation of Plastic, Dayné and Gaston. Messrs Dayné and Gaston founded the company back in 1948. The second of the two gentlemen was originally a barman and bubbling with ideas. For example, how bottles could be re-closed or tied together. Initially he implemented his solutions in wood and cardboard. When plastic became popular after

the war, he decided to change to the novel material and bought a first injection moulding machine. The technique was very rudimentary at the time, which is why the young firm limited itself to moulding closures, capsules and bottle suspension devices. A rack of plastic for bottles was invented in 1962. The company grew slowly and gradually.

PDG Plastiques located in the French town of Malesherbes produces 850 million preforms and 45 million bottles per year.



PDG is now owned by your family. When did you take it over?

My father acquired PDG in 1981, at a time when it was still very small. He had worked for Massilly then, a French producer of tin capsules, and was convinced of being able to make equivalent plastic capsules. The attempt failed, however. He then started to diversify and produced small dishes for ice-cream, cups, closures and lids. Later on, he specialized successfully in thin-walled base cups for PET bottles. Regrettably, this business collapsed from one day to the next with the introduction of the petaloid base.

PDG is now one of the most important domestic PET preform producers. When did this change occur?

Our customers who bought the bottle base cups from us told my father about the unsatisfied demand for preforms. This prompted him to set up the present operation. He bought the necessary machines and entered the PET business. For the first ten years, PDG produced just the preforms, but in the year 2000, we also started to blow bottles.

What further steps followed in the development of PDG?

In 2006, we developed a preform weighing only 2.4 grams and started an innovation programme. This earned us the reputation of a creative enterprise and caused other customers to knock on our door with their ideas and wishes. One project was the preform for milk bottles, for which we won the Oscar of the packaging industry in 2008.

What does PDG look like in numbers?

Right now, PDG employs a workforce of 67. We produce around the clock on seven days per week and on nearly 365 days in the year. Thanks to our highly automated production plant we come to 850 million preforms and 45 million bottles p.a., resulting in total sales to the order of 36 million euros.

Which ones are the most famous brands in your portfolio?

LSDH is an important player in the beverages industry of our country, then the mineral waters of Ogeu, Plancoët, the Inter-marché Group, Andros and Leclerc Group.

PDG produces also wine bottles of PET. What can you tell us about these?

We supply some customers here, but the majority are exported. Thanks to the built-in oxygen barrier the wine in the PET bottles lasts for up to two years, which is an adequate shelf life for wine in most cases. A survey revealed that wine is not stored for longer than two years on average. The PET bottles are quite widely accepted because they hardly vary from the glass bottle at first sight.

Is the PET wine bottle a product with chances for growth?

The market does grow, but only slowly. I guess that it will remain a niche market. I see more potential for beer, with which first tests have already shown positive results. The young generation seems to be more ready to drink beer from nice PET bottles. The chances for a large-scale global spreading of the PET beer bottle seem to be very realistic to me. Something similar – or even more intensively – will happen with the milk bottle.

What reasons do you see for this?

PET, preforms and the blowing technique are the most efficient methods on the market and easy to implement. Mineral waters, soft drinks, juices, but also oil, vinegar and sauces are already all offered in PET containers. I assume that PET will be the most important packaging material in twenty years' time. There's no reason why this development should stop short of the milk. The branch is subject to the same economic factors. Beer and milk are future markets, whereby the milk segment will develop faster.

On which geographical markets do you focus?

France is our main market. It is a mature market in which it



"I SEE MUCH POTENTIAL
FOR BEER AND ESPECIALLY
MILK BOTTLES."

François Desfretier, CEO PDG Plastiques

becomes more and more difficult to achieve growth. At present between five and ten per cent of our annual production are exported. A share which we might expand in the years ahead when we look at the home market. It is quite possible that PDG will intensify its engagement abroad in the future. I cannot even entirely exclude an autonomous production outside France at this stage.



PDG Plastiques S.A.S. of Malesherbes, France, produces preforms and bottles for different segments of the foodstuffs industry. The total annual output is approx. 850 million preforms and 45 million bottles. www.pdg-plastiques.com

You were awarded an Oscar for the 2.4 grams preform. Did any other PDG product win prizes?

We got two Oscars in 2006 for the mini-preform for the children's yoghurt drink. In 2008, our preform with a 100 per

cent light barrier was given an award. It is made using the overmoulding technique and intended for the milk market. Also ground-breaking in our opinion is our pre-blow system, although it hasn't won a prize. It is the result of a cooperation between Otto Hofstetter, Netstal and PDG.

What characterizes the pre-blow preform?

In the past, our industry has made every effort to reduce the bottle's wall thickness in the interest of a low weight. I believe that the minimum is reached. However, we never liked the fact that the base of a standard preform is thicker than the side walls. And we looked for a way to better distribute the material. Seeing that this is not possible during injection, we blow up the tip of the preform by a little while it is still in the mould. Later, when the pre-blow blank is blown to the bottle shape, the plastic material lends itself better to pulling sideways. As a result, the walls become more stable while the preform weight stays the same. An innovation which might never win us an Oscar, but which we feel is at least as pioneering as the 2.4 grams preform.

The ecology is important in your company's philosophy. What efforts do you make in this respect?

Rightly or wrongly, a finger is often pointed at packaging articles made of plastic. Insofar we should try hard to improve the image of these products. Recycling has top priority. No plastic material should go to waste. And there remains much to do in this respect. Especially in France. We should also continue our efforts to further reduce the material consumption.

"THE PREBLOW SYSTEM OPTIMIZES THE WALL SECTION IN THE BASE AREA OF THE BOTTLE AND SAVES WEIGHT."

François Desfretier, CEO PDG Plastiques

Is there a savings potential with the machines?

Yes, progress has already been made with the injection moulding machines, but yet more energy could be saved. The processing of plastic imposes high demands: heating and cooling alone consume a lot of energy and might hold savings potential.

Are bio-based raw materials a topic with PDG?

We watch the development of alternative raw materials very closely. They could in fact be a good option to substitute



Both the spirit of innovation and quality thinking are clearly evident at PDG Plastiques.



Special solutions for exceptional tasks make PDG Plastiques a valuable partner for the foodstuffs industry.

conventional PET. Some products are available on the market, but unfortunately still somewhat expensive. First tests make us confident as regards processing and quality. The 30 per cent bio-based raw materials are absolutely comparable with conventional PET. The future will no doubt lie in this direction, but it will take a little more time.

PDG has been a customer of Otto Hofstetter AG for many years. What are the reasons for your loyalty?

There are two good reasons for our long-standing cooperation: human and technical ones, and they are equally important. Let's start with the technical side. The injection moulds of Otto Hofstetter AG have always convinced us as regards quality, reliability and robustness. We've been producing large series for years and here such characteristics are the strongest technical argument. Then there's the human side, namely the quality of the exchange of ideas and the confidence. We've always been treated as an important customer, right from the beginnings of PDG. And the manner in which problems are dealt with. It never mattered who was to blame if something

didn't work. The professionals of Otto Hofstetter AG always looked for the cause and the best solution. This mentality is the basis of our technical as well as human business relationship and will hopefully remain intact for yet a long time.

What is your vision for PDG in the year 2025?

We shall continue working on our relationship with our customers and suppliers in the spirit of a partnership. This should help our teams to further advance the development of the company. This, however, not at any price. The idea is that the employees of PDG can develop their talents also in future. They spend very much time in the company and have every right to progress personally. I'm hoping for decisive impulses from the internationalization. If we approach all pending projects as a team, we can become a respected and strong actor in the field of preform and bottle production. We certainly have the potential for it.

We, too, are quite convinced of this. Many thanks for this interview, Mr Desfretier.

BUILT ON BOTTLES.

They're more stable than buildings of concrete. Earthquakes, fire and even bullets won't harm them. They keep the room temperature constant at a pleasant 18°C even in hot regions: PET houses, made from PET bottles. One prototype is currently being built in Nigeria. More are due to follow. They'll solve environmental problems there and ease the serious housing shortage. This isn't a fairy tale, even if it reads like one.

In Nigeria the building stones of the future lie on the road by the millions. Used, thoughtlessly discarded PET bottles. They're now recycled for PET-bottle home construction. A first such PET house is now being built near Kaduna in Nigeria's north. The project was initiated by the non-governmental organization DARE (Development Association for Renewable Energies) with the support of specialists from London. Further projects are planned, e.g. the extension of a school with PET.

Unbelievable properties.

The plastic-bottle house is a miracle in many ways. At first sight it appears to be a normal home. But instead of stones, it consists of plastic bottles filled with sand and screwed tight. These "bricks" are placed on top of each other and secured with soil or cob mortar. Also suitable are mixtures from lime and sand or lime and clay soil. As little cement as possible is used. A net of strings connects the bottle necks and gives the

construction extra stability. The project management claims that the bottles filled with sand are more stable than normal concrete blocks.

For the environment; against bullets, the heat and the housing shortage.

"The PET-bottle house is fire-, bullet- and earthquake-resistant. The temperature inside it is a constant 18°C, very good in the tropical climate," a newspaper report quotes Project Manager Yahaya Ahmad. Up to two storeys can be placed on top of the ground floor of the PET-bottle house. The prototype is round and has two bedrooms, a kitchen, an inner court, a bath and a toilet. About 14,000 PET bottles were needed for this building, and Nigeria isn't short of them. On the contrary. If PET-bottle houses are built on a large scale, this can relieve Nigeria in several ways. Africa's most densely populated country produces mountains of waste, also of PET bottles



Houses from discarded PET bottles could solve environmental problems and ease housing shortages in many regions of the world. Some examples – like this one in Nigeria – are already existing.



which are simply thrown away and then build up in roads, canals and streams. Three million a day! The problem disappears of its own accord if they suddenly become a valuable building material. What's more, the country suffers from a severe shortage of accommodation. Sixteen million extra flats are needed. It would take the equivalent of 226 billion euros to build them. With plastic-bottle houses the building costs can be reduced by a factor of four in comparison with conventional homes.

PET bottles become water tanks.

Andreas Froese, inventor of the PET-bottle construction technique, points out that the bottles are also suitable to build tanks for rainwater. Right now he plans PET-bottle projects in 14 countries or has them already under way. For more information: <http://www.eco-tecnologia.com>





THE BRAIN- WORKERS.

Ongoing further development is permanently on the agenda at [Otto Hofstetter AG](#). This is guaranteed by the Department for [Technology & Development](#). In charge of the internal think tank is [Andreas Litscher](#). He and [Stefan Zatti](#) told Inform how the department works, what its priorities are and who benefits from its services.

The Department for Technology & Development – so its official name – is the innovation hub of Otto Hofstetter AG. Here is where the needs and wants of the customers, the market, the field representatives and the strategists meet. They are bundled, discussed with the management depending on the consequences, and led into the right channels.

Tools of the inventor.

Andreas Litscher heads the internal innovation team. He worked his way up within the company. After his apprenticeship as a designer, studies at the Rapperswil College of Technology and an Executive Master in Business Administration he is perfectly equipped for his present job. And he certainly can't complain about a shortage of work in his department.

On the contrary. He is confronted with proposals, wishes and suggestions from all sides. One important task of Andreas Litscher and his team is to decide in an initial phase which development projects to pursue: determining the course, setting priorities.

Stronger by joining forces.

Stefan Zatti, member of the management, explains: "The daily business enjoys top priority in our technology department. If we launch a new generation of moulds on the market, this has absolute priority in our development work." In the area of PET the designers and technicians of Otto Hofstetter AG cooperate closely with mechanical engineers. A regular exchange takes place particularly with the specialists of



Novel solutions for customers come up almost daily in the Technology & Development Department of Otto Hofstetter AG.

Netstal-Maschinen AG. This transfer of know-how is to the benefit of all. Together with the developers of this traditional Swiss enterprise we discuss new machines, machine components, moulds, processes and improvements. And many of the ideas are implemented. In a joint effort the two companies set trends.

Pressing on.

Once the daily business is attended to, the developers can devote themselves to the creation of completely new strategic products. Some of the work runs in parallel. Tubes are an example for this. An entirely new field with which Otto Hofstetter AG makes its technology available to other branches of the industry. Tubes fit well in the company's development strategy and represent a diversification. The relevant innovation process is right at the top of Andreas Litscher's list of priorities. Otto Hofstetter AG intends to set benchmarks with moulds for the production of tubes.

To develop new moulds for the packaging segment, on the other hand, is daily business. Every customer has different packaging products. For these it is a matter of course to adjust the moulds on an ongoing basis and individually, to develop them further, to reinvent them. This applies to both segments, PET as well as PAC.

Ideas from the intermediate worlds.

In the initial stages of the enterprise Otto Hofstetter Senior attended to the T&D work personally at the same time. While he managed production, he was also his own inventor, engineer and designer. The company grew rapidly. Sooner than expected there came the need for an own technical office, today's Department for Technology & Development headed by Andreas Litscher.

His department is behind such innovations as the co-injection moulds for the production of preforms for milk bottles and, on the packaging side, the mould for caviar tubs with odour and

taste barrier. On the programme right now is the development of the overmoulding technique for preforms.

How does it actually happen? Like the mould production at Otto Hofstetter AG, the Department for Technology & Development is divided in a PET and a PAC segment. The multi-layer technique was originally developed for PET bottles and is now applied with great success all over the world. When a long-standing customer expressed the need for a super-light caviar container of neutral odour and taste, the developers of the PET innovation remembered their earlier work and adapted the technique for the new task on hand.

"OUR NEW DEVELOPMENTS ARE NORMALLY COMPATIBLE WITH OLD SYSTEMS."

Andreas Litscher, Department Manager Technology & Development

Backwards ahead of the others.

Completely new developments are rather the exception. Quite often a further development of existing moulds is concerned. And here Andreas Litscher and Stefan Zatti fully agree that Otto Hofstetter AG holds a special position. The magic word: "backwards compatibility".

Stefan Zatti in this context: "You have to see it this way: at Hofstetter, figuratively speaking, the latest VW motor still fits in the traditional VW beetle. With novelties we've always taken care that they can also be fitted in existing moulds, i.e. that they're compatible. This is always a balancing act, of course, and sometimes we're very tempted to sacrifice a few bolts here and there to be able to present the pioneering innovation." But usually the insight wins again that, apart from the quality and the long useful life, this continuous compatibility is a major advantage of the moulds of Otto Hofstetter AG. If the customer desires it.

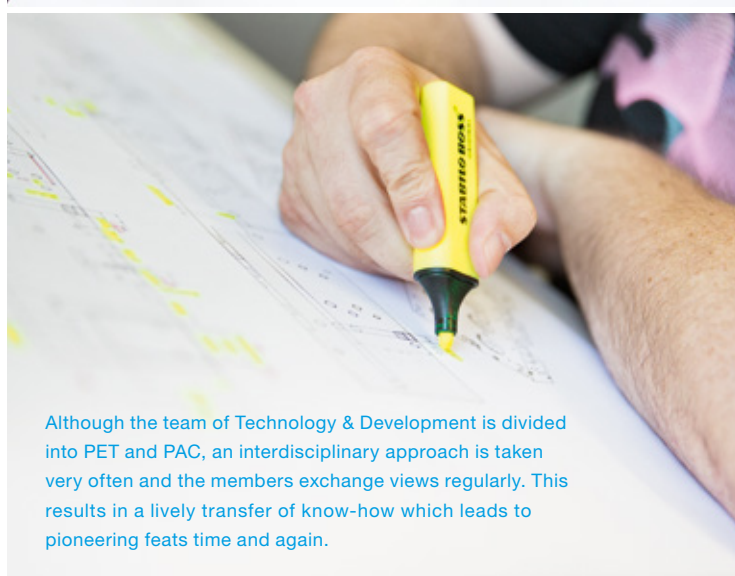
This is a big plus for Otto Hofstetter AG, and an attitude which is pursued relentlessly by the team around Andreas Litscher. Also much to the customers' satisfaction, because this mentality assures them of spare parts being available for many years after they invested in moulds from Uznach. And the environment benefits from the inventiveness as well, since the developers and engineers of Otto Hofstetter AG devote considerable thought also to the ecological footprint of the production. Quite objectively, but not less intensively.



Using grey mass to save grey energy.

Stefan Zatti mentions the development meeting at which the Management and the T&D Department exchange views at regular intervals. In combination with the qualifications and the experience which characterize Andreas Litscher's team the meeting is no doubt an important contributor towards the fact that the innovation projects normally achieve the aims very precisely. It's very rare that a project fails completely. This saves valuable resources.

Andreas Litscher emphasizes that, when it comes to the energy consumption, not only the moulds, but all the other components of the overall system make a decisive contribution. The close cooperation between the experts of the Technology & Development Department of Otto Hofstetter AG and the different mechanical engineering companies pays direct dividends for the environment, and thereby also for the customer. Perfectly fitting moulds help the moulding machines to save valuable energy.



Although the team of Technology & Development is divided into PET and PAC, an interdisciplinary approach is taken very often and the members exchange views regularly. This results in a lively transfer of know-how which leads to pioneering feats time and again.

"THE EXCHANGE OF
EXPERIENCES IS THE PREREQUISITE
FOR INNOVATIONS."

Andreas Litscher, Department Manager Technology & Development





Otto Hofstetter AG



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